银湾研究院（GreyBay Institute）
Great Bay Area Innovation and Research Center

Risk Factors of Underweight and Interventions to Prevent or Mitigate Underweight in Young Adult Males in Japan: An Evidence-based Research Project for China

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Abstract

Underweight and related eating problems among young men have received far less attention than those among young women, despite growing evidence that body dissatisfaction, disordered eating, and unhealthy weight-control behaviors are increasingly present in male populations. In Japan and other East Asian settings, these risks may be shaped not only by thinness-related pressures, but also by muscularity-oriented ideals, masculine role expectations, and emerging social and media influences. This report, produced by the GreyBay Institute, reviews the evidence on risk factors and interventions related to underweight among young adult males in Japan, with the aim of drawing lessons relevant to the Chinese context.

Based on the literature, the report develops an ecological framework that organizes determinants into social and cultural, family, and individual levels. At the individual level, psychological, behavioral, sociodemographic, and physiological factors are examined in detail. The evidence suggests that underweight and disordered eating in young men arise through a complex pathway involving sociocultural pressure, body image disturbance, internalization of muscular or lean ideals, emotional distress, and weight-control behaviors. Family environment, parental education, and exposure to family violence also appear to play important roles, while physiological factors such as BMI, genetics, and pre-existing health conditions may directly influence risk.

The report further reviews available intervention evidence, which remains limited but provides useful insights. Existing studies suggest that earlier detection by clinicians and educational institutions, self-esteem–focused school programs, improved awareness of male-specific symptoms, and interventions that help young men critically examine appearance ideals may all be promising directions. Drawing on Japan's experience, this report offers an evidence-based foundation for future research and policy development in China, particularly for designing gender-sensitive strategies to prevent underweight, improve body image, and support the health of young men.

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1. Background

Eating problems, anorexia nervosa, bulimia nervosa, eating disorders not otherwise specified (EDNOS), and disordered eating, have been studied extensively among adolescent girls, young women, and college women.[1] The few existing studies of eating problems in young men suggest that eating disorders are less prevalent among men than among women.[2] A recent study found that men were more likely than women to have a later onset of eating disorder (20.6 y vs 17.2 y, respectively)[3]. Studies of adolescents have found that significant numbers of boys (5%-20%) report restrained eating, vomiting, laxative abuse, or smoking cigarettes for weight control. Furthermore, the data showed that men constituted an increasing percentage of body image concerns and eating disorder admissions, approximately 30% of males report a desire to change their body size or shape [4], suggesting that eating problems may be increasing among young men. Body image concern is defined as overvaluation of appearance, body dissatisfaction, and/or a distorted view of our body and have long been associated with weight and eating related problems [5].

Actually, public and scientific interest in men with eating disorders continues to grow. Research suggests that men account for up to 33% of cases of anorexia and bulimia nervosa diagnoses [6] and up to 90% of cases of muscle dysmorphia [7]. Moreover, it has become increasingly clear that gender differences qualify the experience of eating disorders. For example, the ideal body coveted by males tends to be overtly muscular in comparison to the thin body coveted by most females [8]. Thus, to accommodate the “male experience” of eating disorders, researchers have developed novel measures of men’s eating disorder symptomatology [9], proposed a reconceptualization of disordered eating to include muscularity-oriented eating attitudes and behaviors [10], and proposed that muscle dysmorphia be reclassified from a subtype of body dysmorphic disorder to an eating disorder [11]. Gender role norms, described as the set of attitudes, behaviors, and thoughts that societal forces deem as representative of the male or female gender [12], may be crucial to understanding the

male experience of eating disorders. Traditional masculine norms include dominance, power, status, confidence, sexual success, and the exercise of physical and emotional self-control [12], and adherence to these norms is reinforced and deviation punished via peers, family and the media [13].

The masculinity hypothesis [14] proposes that conformity to masculine gender roles is a risk factor for muscularity-oriented body image pathology in males. Having an overtly muscular body that is characterized by well-developed muscle mass and low body fat may be perceived as a means by which men can communicate traditionally masculine qualities to others. Men who feel that they lack these masculine qualities, or who wish to project these qualities, may then feel pressured to obtain a larger, more muscular body, which in turn may lead to body dissatisfaction and disordered eating. The masculinity hypothesis is closely related to the theory of threatened masculinity, which proposes that formerly stable pathways to achieving a masculine identity have become destabilized because of increasing parity between the sexes, and that men are building muscles in increasing numbers in an attempt to reclaim and communicate their masculinity [15].

Evidence in support of the masculinity hypothesis is accumulating. Greater adherence to masculine norms has been linked to increased muscle dissatisfaction [14] and increased drive for muscularity [16][17]. Men with muscle dysmorphia exhibit greater conformity to masculine norms than healthy control men [18]. Young men rate people who exhibit symptoms of muscle dysmorphia symptoms as more masculine than people who exhibit symptoms of anorexia nervosa [19], and it has been suggested that the desire to appear or be more masculine may explain the positive association between young men's disordered eating and their desire to be like someone with muscle dysmorphia. However, despite growing support for the masculinity hypothesis, a recent comprehensive review of the literature by Blashill (2011) highlighted several shortcomings [14]. First, most research has used measurements of gender roles based exclusively on personality traits, whereas newer multidimensional instruments, such as the Conformity to Masculine Norms Inventory [16] and Conformity to Feminine Norms Inventory [20] assess a broader definition of gender roles that includes gender-

normative behaviors, attitudes, and personality traits. Few studies have used these instruments in the context of males' body image and eating concerns. Second, researchers have focused their efforts on muscle dissatisfaction and neglected body fat dissatisfaction, despite the latter being a distinct and equally important component of male body image [21][14][22].

A third criticism made by Blashill (2011) is that research examining the links between gender roles, body image and eating among men has neglected femininity. Feminine gender norms include niceness, passivity, deference to others and interpersonal dependence, and these are thought to cultivate lowered self-esteem and a tendency to seek approval from others [23]. Low self-esteem and approval-seeking behavior may motivate the pursuit of a body that is perceived as ideal and desirable, which in turn may result in body dissatisfaction, disordered eating, and eating disorders [23]. Thus, the "femininity hypothesis" proposes that conformity to feminine norms is a risk factor for eating pathology and body dissatisfaction among women and has received considerable empirical support [23][24]. Few studies have explored whether the femininity hypothesis generalizes to men by examining the relationship between multi-dimensional constructs of femininity and disordered eating in male populations. Like women, men who report greater adherence to feminine norms may try to win approval from others by pursuing what what they perceive to be a desirable and ideal male body. Some support for this hypothesis comes from [18] Murray et al. (2013), who found that conformity to feminine norms was elevated in men with anorexia nervosa compared to healthy control men. Interestingly, femininity was not elevated in men with muscle dysmorphia [18], suggesting that the femininity hypothesis may generalize to men only insofar as thinness-oriented disordered eating is concerned. In opposition to this argument, the review by Blashill (2011) did not find a significant relationship between femininity and body fat dissatisfaction in men, or between femininity and eating pathology. Blashill further noted, however, that an insufficient number of studies using multidimensional measures of femininity precluded a specific analysis of the relationship between multidimensional measures of femininity and body image and eating pathology. In addition to the shortcomings mentioned by Blashill

(2011), research on gender roles, body image, and eating have neglected muscularity-oriented disordered eating. A pathological desire to increase muscularity may entail a pattern of disordered eating behaviors and attitudes that are geared toward the acquisition of muscle mass, conceptualized elsewhere as "muscularity-oriented disordered eating" [10][25]. In support of this distinction, men diagnosed with anorexia nervosa report greater thinness-oriented disordered eating and less muscularity-oriented disordered eating than men diagnosed with muscle dysmorphia [25]. The finding that men with muscle dysmorphia also demonstrate greater conformity to masculine norms and less conformity to feminine norms than men with anorexia nervosa [18] suggests that thinness- and muscularity-disordered eating may be differentially associated with conformity to masculine and feminine gender roles.

In addition to the conformity to masculine norms which predict increased muscle dissatisfaction and muscularity-oriented disordered eating, underweight issues among males are also associated with emotional distress, low sense of self-value and other factors [26][27][28]. Many factors have been implicated in the onset and maintenance of body image problems, including physiological factors (e.g., body mass index), psychological factors (e.g., perfectionism, negative affect, low self-esteem) and sociocultural factors (e.g., family, peer, romantic partner and media) [29]. These factors play a role in weight issues among males from their early adolescence to mid-late adolescence. For example, the transformation from childhood to adulthood is characterized by rapid physical growth, large increases in hormone levels and the appearance of secondary sexual characteristics. During the pubertal development at their early adolescence, their nutritional status may be influenced by the many factors affecting pubertal development such as physical growth, peer group pressure, environmental factors (e.g. the media) and psychological factors (e.g. body image).

Based on the above wide-ranging review of research literature, we can see that dissatisfaction with body size and shape is a common concern for adolescents and young male adults. This report aimed to examine the associations between risk factors (sociocultural level, family level, and individual level), body dissatisfaction, and disordered eating in males.

2. Risk factors of underweight

2.1 Overview

Based on the literature reviews in this project, we classified all factors that cause underweight in young men into three levels, social and cultural level, family level, and individual level. For individual level, the factors were further separated into four aspects, namely, psychological, behavioral, sociodemographic and physiological factors.

Figure 2-1 describes the hierarchy of the factors related to underweight among young men. Detailed factors are provided under each big category and will be further discussed in the next three subsections in accordance with the three levels.

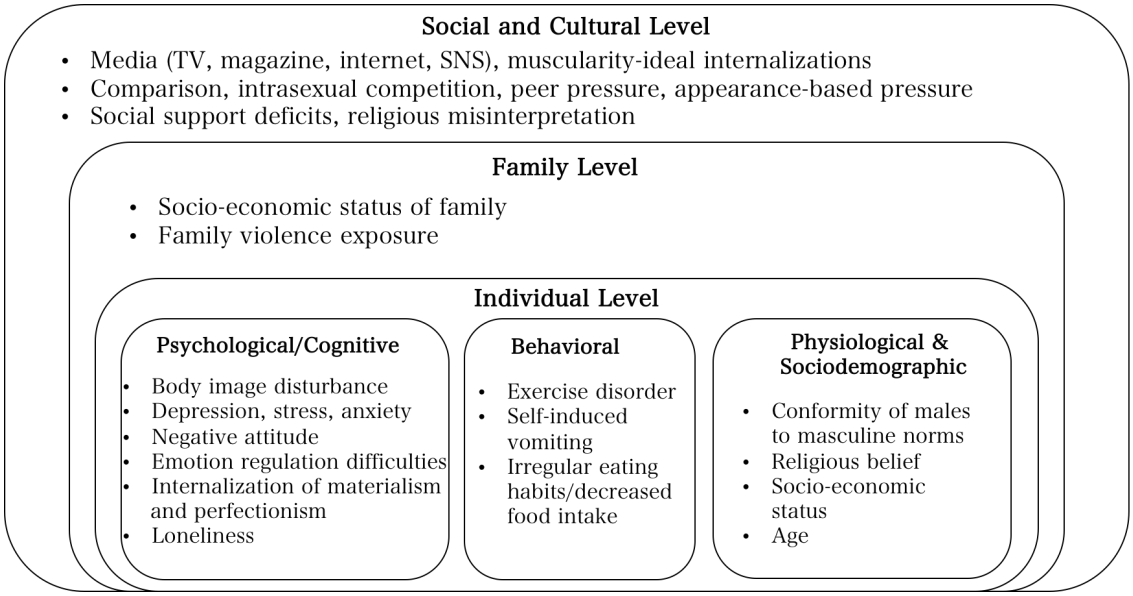

Fig. 2-1 An Ecological model for the hierarchy of factors that cause underweight in young men

While in Figure 2-2, the logic web presents the hypothesized causal relationship between each factor, which was summarized from all review literatures. The links on the figure are not all necessarily causation relationship due to the limitations of study designs. Some of the links in this logic web might need to be further studied, but this at least helps to understand the underlying relationships between the complex factors.

What needs to be noticed here is that body image disturbance/concern, theoretically, is within the category of psychological and cognitive status. But decent amount of psychological factors, such as poor emotion regulation, are associated with body image disturbance. Thus, body image disturbance is separated into a tier here. In Fig. 2-2, the individual level is marked within the blue dash line.

Evidence that support each of the arrows are discussed in the corresponding subsections.

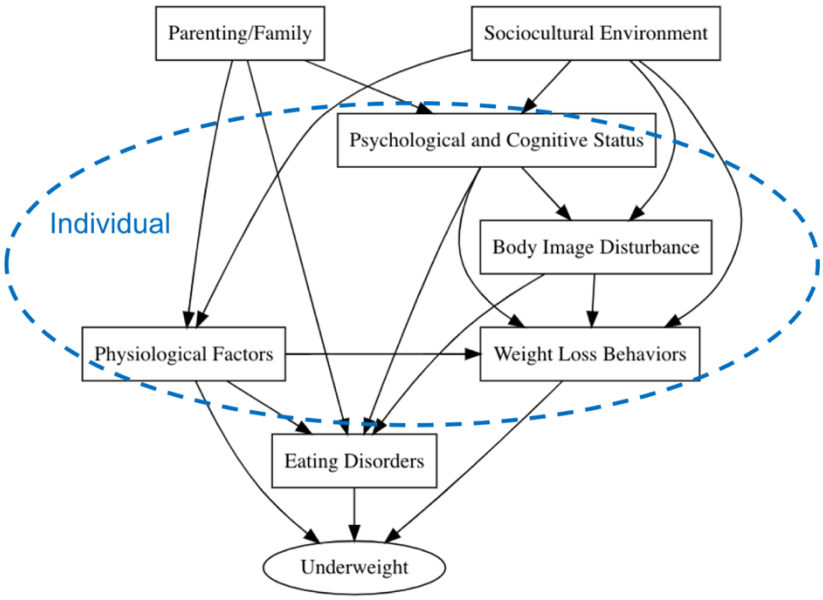

Fig. 2-2 The underlying logic web that lead to underweight in young women

2.2 Social and cultural level

Sociocultural factors (i.e., appearance-based social pressure, teasing, comparison, media, social support, culture, and socio-economic status) have received increasing attention in leading body image disturbances (BID) and weight loss behaviors. These factors could also cause psychological and cognitive bias such as internalization of thinness, leanness, or muscularity for men. Not only adolescent males but also men who are in their late adolescence are affected by ranges of social factors. For example, despite reduced adherence to adolescent peer group norms by late adolescence, effects of appearance teasing and pressure may persist into adulthood. [30]

Through influencing emotional, psychological, and cognitive status, social and cultural environment serves as main external stimuli for changing people's perception of their body weight. Appearance pressure, teasing, and comparison may be salient influences on a range of body image concerns within diverse adolescent and young adults. [31] Pressure from social media, peers, sport involvement, socially prescribed perfectionism, and a history of weight teasing lead to body dissatisfaction. [32][33] What is more, socio-economic status drives weight issues.[34]

Sometimes, perception of teasing and social pressure directly contribute to body dissatisfaction for males, not only in Western cultures but also in Eastern cultures. However, collectivistic cultures such as China, many people have a tight and close personal network, including parents, romantic partner, family, intimate friends and some other relationships, could moderate the relationship between social support and body dissatisfaction. [27]

In addition to pressure from social media and peers, some religion beliefs also tend to drive decreased food intakes. For example, a paper presents three cases involving young Orthodox Jewish males, each of whom lost 15-25 pounds over a course of time ranging between a few months and up to 2 years, as the result of decreased food intake because of misinterpretation of a religious concept learned in their Judaic studies.[35]

Body image disturbance and body dissatisfaction is also related to assortative mating (and regional clustering) since it is associated with the heritability of BMI. [36]

2.3 Family level

Linear regression analyses showed that high paternal education was inversely associated with a lower BMI ($P = 0.035$) and the concentration of total cholesterol ($P = 0.003$) and low-density lipoprotein (LDL) ($P = 0.014$) among young adult men. [37] One reason is that families of higher socioeconomic strata (income and education of parents) were more devoted to children's sports activities.

In addition to the socio-economic status of families, repeated incidents of family

violence exposure lead to depression and anxiety among young males. Because emotion regulation appears to be of importance to the development and maintenance of disordered eating, direct victimization by family members and witnessing violence in the home is associated with eating subsequent eating disorder symptoms. [38]

2.4 Individual level

2.4.1 Psychological factors

Psychological factors including internalizations, emotion, perception of body image, and attitude lead to underweight indirectly through direct impacting on body image disturbance, weight loss behaviors, and eating disorders.

A study which explores the body image ideals among a racially/ethnically diverse sample of gay and bisexual men in the United States found out that the dominant ideal male body is muscular and thin. [39] Thin-ideal internalization did not significantly relate to body dissatisfaction. One possible reason for this null finding is that the ideal thin portrayed in culture mostly focuses on females' body but not on males'. [27] This proves males' internalization of muscular perfect ideals in a way. Internalization of perfect ideals, materialism values, the muscular/athletic ideal is strongly related to eating restrictions and muscularity enhancement behaviors among young males.[40][41] Within these aspects of internalizations, the internalization of materialistic values is strongly linked to the internalization of body-perfect ideals: the thin-ideal for young women, and the muscular-ideal for young men. [42]

Also, muscularity dissatisfaction was uniquely associated with both emotion regulation difficulties and an attentional bias toward rejecting faces. Body fat dissatisfaction was not uniquely associated with any aspect of emotional functioning.[43] Emotion regulation difficulties and negative attitude towards life, such as depression and anxiety, may play a role in the development and maintenance of eating disorder symptoms and weight issues.[34][38] Among all the emotions, loneliness was a stronger predictor for adolescent males eating disorders, as compared to females.[44]

What is more, body image (place low values on appearance) leads to greater weight dissatisfaction and eating disorders.[6][41] Psychological factors sometime leads to behavioral factors. For instance, body weight dissatisfaction leading to self-induced vomiting and exercise disorder. [14] Also, there is a strong and direct relationship between the internalization of the muscular/athletic ideal and muscularity enhancement behaviors. [41]

2.4.2 Behavioral factors

Excessive weight loss behaviors directly lead to eating disorders among men or even the result underweight. Based on previous studies, weight loss behaviors are likely to be caused by sociocultural factors, psychological and cognitive status, body image disturbance, and other variables such as BMI. For example, there is a strong and direct relationship between muscularity enhancement behaviors and internalization of the muscular/athletic ideal.[41] Weight loss behaviors is also related to body dissatisfaction. For instance, self-induced vomiting and exercise disorder among young men is driven by their dissatisfaction of body weight.[1] Besides that, M. P. McCabe and L. A. Ricciardelli 2001 study found out that boys with a larger body mass index (BAG) were less satisfied with their muscle tone and more likely to change their eating habits to decrease their body size or shape than boys with a smaller BMI. [4] It is also reported that decreased food intake behavior is driven by the misinterpretation of a religious concept learned in males' Judaic studies. [35]

2.4.3 Sociodemographic factors

Sociodemographic factors such as sex, age, ethnicity, and socio-economic status could drive disorder eating and weight issues. In terms of gender role effects, as compared to females whose ideals are merely thin, the dominant ideal male body is muscular and thin. Racial/ethnic minority participants added this ideal is further characterized as white/light-skin toned. [39] Thus, increased conformity of males to masculine norms predicted greater muscle dissatisfaction and muscularity-oriented

disordered eating, but not body fat dissatisfaction or thinness-oriented disordered eating.[43]

In addition to gender roles and ethnicity, socio-economic status of males also lead to weight issues since people with higher education and income tend to have stricter involvement in sport activities and healthier diet which decrease probability of eating disorder.[34] It is possible that they have a more objective perception of body image. Thus, people with higher education and income are more likely to have lower BMI and subsequently less eating disorder behaviors.

2.4.4 Physiological factors

Despite psychological factors greatly lead to body image disturbance which ultimately cause eating disorder and underweight, personal physiological factors such as gene, BMI, anorexia nervosa, and malnutrition usually have directly impacts on eating disorders. Compared to internalization of muscular ideals, these factors directly drive the weight loss behaviors or eating disorder because it is usually not under control. For example, in a study of young men in Sweden by Magnusson PK. et al., the familial risk of having BMI values above or below various cut-offs was found to be equally strong for low BMI as for high BMI values. The almost twice as strong BMI correlation between maternal half-brothers as between paternal half-brothers illustrates the importance of factors of non-additive genetic origin, to the familial aggregation of BMI. The significant BMI association found between biologically unrelated individuals from the same family emphasizes that the heritability of BMI is closely related to assortative mating (and regional clustering). [36]

In addition to genetic effects on BMI, BMI is found to be inversely associated with illicit drug use by males after adjusting for possible confounding factors such as level of education, nicotine dependence, breath carbon monoxide (CO) levels, and alcohol abuse and dependence.[45] Boys with a larger body mass index (BAG) were less satisfied with their muscle tone and more likely to change their eating habits to decrease their body size or shape than boys with a smaller BMI.[4]

There are other conditions other than BMI would lead to eating disorder and

underweight directly such as anorexia nervosa and significant abnormalities and malnutrition. [46] [47]

3. Interventions to Prevent or Mitigate Underweight

3.1 Overview

Based on the literature reviews, there are totally four studies recognized as intervention studies. Although these four studies are not strictly randomized control experiment to verify the intervention effects, but they still provide significant insight to current project, so we decided to include them as intervention studies.

Interventions can also be classified into the three levels (social, family, and individual), according to the ecological model. Following the model, individual level interventions could be further classified to psychological, behavioral interventions (sociodemographic and physiological factors belong to individual properties and are not changeable).

The social level intervention mainly includes that clinicians, health professionals, health services should be more involved to detect the signs and symptoms of changes in weight and excessive exercise which predict eating disorders in male children (2-1, 2-4). Support from society and families are encouraged to reduce eating disorder through improving self-esteem (2-3). From the perspective of individual level intervention, males are encouraged to participate in critiquing the appearance ideals through written and verbal exercises, in order to reduce their internalization of the ideal and improve their body image (2-2).

3.2 List of successful interventions

2-1 Clinical presentation of eating disorders in young males at a tertiary setting.			
Summary	Clinical presentation of eating disorders in young males implies better intervention.		
Country	Australia	Year	2015
Goal	Provide knowledge about the clinical presentation of eating disorders in young males.		

Study Design	Prospective cohort study
Participants	Children and adolescents aged 8 to 17 years who were consecutively assessed by the Child and Adolescent Mental Health Service (CAMHS) Specialized Eating Disorders Program. The sample consisted of 757 consecutive referrals; n = 53 young males with eating disorders, n = 704 young females with eating disorders.
Background	The majority of individuals with eating disorders are female and this has generated female-centric research, with few studies examining clinical presentation among males. There is a pressing need for research on clinical presentation in males especially among boys.
Methods	The data source was the Helping to Outline Pediatric Eating Disorders (HOPE) Project (N ~ 1000), a prospective, ongoing registry comprising consecutive pediatric (<18 years) tertiary eating disorder referrals. Young males with DSM-5 eating disorders (n = 53) were compared with young females with eating disorders (n = 704).
Results	● There was no significant difference in the prevalence of diagnosis of bulimia nervosa (2 % vs 11 %, p = 0.26) among sexes. ● Males had comparable duration of illness (9 months; p = 0.28) and a significantly earlier age of onset (M = 12 years; p <0.001). Shape concern (2.39 vs 3.57, p <0.001) and weight concern (1.97 vs 3.09, p <0.001) were lower in males, and body mass index z score (−1.61 vs −1.42, p = 0.29) and medical compromise (odds ratio [OR] = 0.64, 95 % CI: 0.36, 1.12) were comparable. ● Males had two-folder higher odds of being diagnosed with unspecified feeding or eating disorders (40 % vs 22 % for females, p = 0.004). Driven exercise to control weight and shape was common and comparable in prevalence among males and females (51 % vs 47 %, p = 0.79) and males were less likely to present with self-induced vomiting (OR = 0.23, 95 % CI: 0.09, 0.59).
Conclusions	● Boys with eating disorders are an understudied group with similarities and differences in clinical presentation from girls with eating disorders. ● Parents and physicians are encouraged to consider changes in weight, disturbed vital signs, and driven, frequent exercise for the purposes of controlling weight or shape, as possible signs of eating disorders among male children. ● Diagnostic classification, assessment instruments, conceptualization, and treatment methods need to be refined to improve application to young males.
Link to the Fukuoka Project	● Clinicians and parents in Fukuoka city should be alerted to the signs and symptoms of changes in weight, unstable vital signs, and driven exercise for the purposes of controlling weight and shape, as means to identify eating disorders in male children. ● Clinicians in Fukuoka city should refine assessment instruments,

	diagnostic classification, and treatment applications to enhance applicability to males.
Reference	CY., S., K., L., C., H., J., M., KJ., H., MJ., H., & HJ., W. (2015). Clinical presentation of eating disorders in young males at a tertiary setting. *Journal of Eating Disorders*, *3*(PG-39), 39. https://doi.org/10.1186/s40337-015-0075-x

2-2 Young men's minimization of their body dissatisfaction.			
Summary	Alternative approaches to evaluate men's minimization of their body dissatisfaction.		
Country	The U.K.	Year	2018
Goal	To examine men's body dissatisfaction qualitatively		
Study Design	Prospective cohort study		
Participants	Forty-two men enrolled in psychology degrees at two universities in the South- and North-West of the United Kingdom participated in the intervention. Participants were aged between 18 and 45 years (M=21.02, SD =5.21). The majority of the sample identified as heterosexual (n=39; 93%), with few participants identifying as sexual minorities (e.g. gay, bisexual; n=3, 6.9%). Most participants identified as British (n¼40; 96%) and White (n=33; 84%).		
Background	Men's body dissatisfaction is now so prevalent as to be normative. As a result, researchers are increasingly directing their attention towards understanding the impact of body dissatisfaction on men's wellbeing, and to developing and evaluating interventions to support men with body dissatisfaction		
Methods	Forty-two British men aged 18-45 years took part in a two-session group intervention across 12 groups. The intervention was designed to improve body dissatisfaction by engaging them in a critique of the appearance ideal through written and behavioral exercises.		
Results	● Men minimized the existence of their own body dissatisfaction while (somewhat surprisingly) outlining the ubiquity and potency of the appearance ideal for men in general. ● Involved men reporting the problematic impact of body dissatisfaction in their lives (despite earlier minimization), such as social avoidance, strict eating and supplement regimes, or difficulty in situations where the body was exposed.		
Conclusions	The results stress the need to acknowledge that men experience a range of impacts of body dissatisfaction beyond clinical presentations (such as disordered eating) that influence their everyday lives, while also recognizing that they tend to minimize this dissatisfaction in conversation.		
Link to the Fukuoka Project	● Although this finding was based on the British men and aimed to describe body dissatisfaction qualitatively, it has important implications for advocacy and interventions to improve men's body		

	dissatisfaction. ● Participating in written, verbal, and behavioral exercises that prompt them to critique the appearance ideal, in order to putatively reduce their internalization of the ideal and improve their body image.
Reference	Jankowski, G. S., Gough, B., Fawkner, H., Halliwell, E., & Diedrichs, P. C. (2018). Young men's minimisation of their body dissatisfaction. *Psychology and Health*, *33*(11), 1343–1363. https://doi.org/10.1080/08870446.2018.1496251

2-3 Improving the body image, eating attitudes, and behaviors of young male and female adolescents: A new educational approach that focuses on self- esteem.			
Summary	Improving the Body Image, Eating Attitudes, and Behaviors of Young Male and Female Adolescents by focusing on self-esteem		
Country	Australia	Year	2000
Goal	the effect of an interactive, school-based, self-esteem education program on the body image and eating attitudes and behaviors of young male and female adolescents following the program and after 12 months		
Study Design	Randomized control experiment		
Participants	A total of 470 secondary students, 173 males (37%) 297 females (63%), aged 11.1–14.5 years, participated in the study. All students who were enrolled in Years 7 and 8 at the schools participated in the study.		
Background	The school environment is a suitable place to implement health promotion programs because adolescent students are accessible and motivated to become involved in educational activities. But previous health promotion programs have negative effects on students.		
Methods	All 470 eligible students (47% male) aged 11–14 years volunteered to participate. The students were randomly allocated to control or intervention groups according to school class groups. The intervention group students participated in the program, whereas the control group students received their scheduled personal development and health class.		
Results	● The program significantly improved the body satisfaction of the intervention students and significantly changed aspects of their self-esteem; social acceptance, physical appearance, and athletic ability became less important for the intervention students and more important for control students. ● One year after the intervention, body image and attitude changes were still present.		
Conclusions	The current study was successful in changing students' body satisfaction, body image, attitudes to eating and body weight, and weight losing behavior because it focused on producing positive changes in the students' self-perceptions and values. This new educational approach ignored attempts to change knowledge.		

Link to the Fukuoka Project	● Fukuoka city could improve body image and eating behaviors by employing cooperative, interactive, and student-centered approaches that develop self-esteem. ● Avoid the kind of direct instruction about diets and eating disorders that may have the potential to result in the adoption of unhealthful eating and dieting behaviors.
Reference	O'Dea, J. A., & Abraham, S. (2000). Improving the body image, eating attitudes, and behaviors of young male and female adolescents: A new educational approach that focuses on self- esteem. *International Journal of Eating Disorders*. https://doi.org/10.1002/(SICI)1098-108X(200007)28:1<43::AID-EAT6>3.0.CO;2-D

2-4 Eating and exercise disorders in young college men			
Summary	Screening could help identification of eating disorder among college men.		
Country	Australia	Year	2002
Goal	Investigate disordered eating and eating and exercise disorders in a male college population		
Study Design	Participants completed the Eating and Exercise Examination (EEE-C) in a computer lab. An eating disorders specialist conducted clinical examinations using the Eating Disorder Examination (EDE) to validate the EEE-C diagnoses of eating disorders.		
Participants	93% undergraduate college men from randomly selected class groups in the faculties of education, science, and medicine volunteered.		
Background	Men constituted an increasing percentage of eating disorder admissions between 1984 and 1997, suggesting that eating problem may be increasing among young men. Significant numbers of boys (5%-20%) report restrained eating, vomiting, laxative abuse, or smoking cigarettes for weight control. The description of the eating disorder problems is usually limited to patients who present for treatment of their clinical eating disorders.		
Methods	Participants completed the Eating and Exercise Examination (EEE-C) in a computer lab. An eating disorders specialist conducted clinical examinations using the Eating Disorder Examination (EDE) to validate the EEE-C diagnoses of eating disorders.		
Results	9% students reported that they had suffered from disordered eating, 18% reported weight control problems, and 22% reported a binge-eating problem. None of them had sought help or been treated for eating disorders or eating problems.		
Conclusions	20% of the young college men in this study displayed eating attitudes and behaviors characteristic of eating disorders and disordered eating.		

Link to the Fukuoka Project	College health professionals should be aware that young men of all shapes and weights may be sensitive about their bodies and that some may undertake extreme weight-control methods. College healthcare services and health education activities should address the weight concerns of young men to make education and treatments most relevant, effective, and available.
Reference	O'Dea, J. A., & Abraham, S. (2002). Eating and exercise disorders in young college men. *Journal of the American College Health Association*,*50*(6),273–278. https://doi.org/10.1080/07448480209603445

4. Findings and conclusions

Based on the literature reviews, we constructed an ecological model which includes include three levels: social and cultural level, family level, and individual level, hierarchically. The underlying causal relationship and associations between the detailed risk factors are represented through a logic web. An overall hierarchy of the logic web, in accordance with the ecological model, is that it constitutes (1) social and cultural level – sociocultural environment; (2) family level – parenting/family; (3) individual level – psychological and cognitive status, body image disturbance, weight loss behaviors, physiological factors; (4) outcome/result – eating disorder and underweight.

Social and cultural status could be considered as the fundamental stimuli for not only psychological and cognitive status but also parenting/family among males. It is human nature to live in groups and receive all the influence from the surrounding environment. Thus, it is of no surprise that these two factors play essential roles in driving body image disturbance and weight issues. From the perspective of social factors, pressure from peers, teasing, and media cause the internalization of thin ideals among males, and effects of appearance teasing may be carried into late adulthood. Lack of social support from society and families increases the probability of body dissatisfaction and eating disorders. For example, Chinese culture tend to have closer relationship with surrounding communities which significantly support mental

appreciation of their own identity including body image. Different from females' early internalization of materialism and perfectionism, males tend to have later onset of body dissatisfaction. What is more, females have higher perception and preference of thin ideals, however, males have increased conformity to masculine norms. This leads to muscularity-oriented body image and weight loss behaviors rather than thinness- and leanness-oriented weight loss behaviors. Because muscularity-oriented weight loss requires different methods from normal to achieve the goal, some males take risk to use illicit drugs and result in rapid decrease in BMI. An interesting finding is that misinterpretation of religions also causes decreased food intake behaviors among males.

From the aspect of family factor, higher parental education level is associated with lower BMI among males. But history of family violence exposure lead to depression and anxiety which is related to the subsequent eating disorder symptoms. Findings at this level are in line with those of females. Individual level risk factors have a greater variety. From the psychological aspect, internalization of perfect ideals, materialism values, the muscular/athletic ideal is strongly related to eating restrictions and muscularity enhancement behaviors among young males.[40][41] Emotion regulation difficulties and negative attitude towards life, may play a role in the development and maintenance of eating disorder symptoms and weight issues.[34][38] Interestingly, among all the emotions, loneliness was a stronger predictor for adolescent males eating disorders, as compared to females.[44] This provides an valuable implication towards intervention. Because of body dissatisfaction, males implement some weight loss behavior such as self-induced vomiting and exercise disorder. Notably, in a males' Judaic studies, participants had decreased food intake behavior due to the misinterpretation of a religious concept. [35] Socio-economic status is associated with weight issues because males with lower education and lower income tend to have poor emotion and anxiety, less involvement in sport activities, and unhealthy dieting. This category of males is more likely to have higher BMI and eating disorders. What is more, personal physiological factors such as gene, BMI, anorexia nervosa, and malnutrition usually have direct impact on eating disorders. And the heritability of BMI is closely

related to assortative mating (and regional clustering). [36]

There is no large number of interventions found during the literature reviews. But the four interventions included provide significant insights into mitigating underweight issues among males. At the social and familial level, the clinicians, health professionals, health services in different communities should increase detection of weight change symptoms, and provide sufficient social support through improving males' self-esteem. For individual intervention, males could increase their participations in written and verbal communication with the communities and express their thoughts about body image concerns. Not restrictive to the interventions included, all the detailed risk factors and underlying association with eating disorders and underweight cast light on potential multidimensional interventions. For instance, males with lower socio-economic status or males from families with lower parental educations could be targeted population for intervention implement. In short, further research about evidential intervention is still demanded.

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